Signal Dojo

Everything you need to learn with it and run it, on one page.
Offline. Needs only `python3`. Runs on a Raspberry Pi, Linux, macOS, or Windows (WSL).

Start it

```
tar -xzf pi-sdr-academy.tar.gz && cd pi-sdr-academy
./run.sh                  # learner
ACADEMY_ADMIN=1 ./run.sh # admin (password)
```

Open `http://127.0.0.1:8080/`. Change it with `HOST=` / `PORT=`. No pip, no internet. Stop with **Ctrl-C**.

Windows: in WSL (Ubuntu), unpacked in `~`, not `C:\`. See `docs/WSL.md`.

Install once (lab machine)

```
sudo apt install vim tmux curl gdb \
  build-essential cmake pkg-config \
  man-db manpages-dev tree net-tools
```

Windows laptops: `wsl/setup-ubuntu.sh` installs these inside WSL.

LEARNER

1 The loop for every challenge

1. **Enter the lab**, pick the next unlocked challenge on your belt.
2. **Read lesson** first: the idea, new words, further reading.
3. **Start**: files are copied to your folder, `~/signal-dojo/<username>`, and a terminal opens there.
4. Read `README.md` (the tasks), edit the files in **vim**.
5. Run `./check` until it prints `CHECK_OK`.
6. Press **Mark as done**, then **Next challenge**.

Start refills the folder; your files come back when you return. **Restart** gives clean ones.

2 Belts

Belts unlock in order: solve every challenge on one to open the next.

White · terminal | Yellow · Linux | Orange · C++ | Blue · Python

3 When you are stuck

- **Hint** reveals the next hint, one level at a time. Hints used are recorded on your profile, so read the lesson again first.
- The `LESSON.md` in your folder is the same lesson the app shows.
- `./check` says which task failed and why. Fix one task at a time.

4 vim survival kit

`vim tasks.py`	open a file
`i` `Esc`	start typing / stop typing
`:w` `:q` `:wq`	save / quit / save and quit
`:q!`	quit and throw changes away
`u` `/word`	undo / search

vim is the only editor taught; `git commit` and `crontab -e` open it too.

5 From the terminal

`./academy next`	stage the next unlocked challenge
`./academy hint`	reveal the next hint
`./academy submit`	run `./check` and score it
`./academy status`	your scoreboard
`./academy save`	progress + your scripts in one .zip (`restore FILE` brings it back)
`cd ~/signal-dojo/<username>`	jump to your challenge folder

In `~/pi-sdr-academy`, with `ACADEMY_USER=<username>` in front.

6 Profile and look

Save to a file in the lab keeps your progress and scripts in one .zip; **Restore** brings them back on any machine. **Profile** shows your belts, solved challenges and recent activity. The round button at top right switches dark and light.

ADMIN

1 Admin password

- The first `./run.sh` asks you to set it; every admin run asks for it in the terminal.
- Change: `./academy admin --change-password`
- Forgot: `./academy admin --reset-password`; the next admin run sets a new one.
- Only a salted hash is kept (`.academy-data/admin.json`). It keeps learners out, not someone who can edit your files.

2 Open any challenge

- In admin mode every belt is unlocked. In the lab, the **Admin mode** box jumps to any challenge by id or title.
- `/admin` lists every challenge with a filter, and **Launch any challenge**.
- Link `/#challenge/<id>`, or terminal `ACADEMY_ADMIN=1 ./academy start <id>`

3 Edit the curriculum at `/admin`

- **+ New challenge**: a 6-step wizard that writes a challenge stub under `curriculum/modules/`.
- **Suggest module / dojo change** saves a proposal; **Review proposals** applies it.
- After editing YAML by hand, run `python3 scripts/check_curriculum.py`.

4 Run a classroom

```
./academy serve --host 192.168.1.50 --no-signup
./academy user add alice      # create account
./academy user passwd alice   # reset password
```

- Learners browse to `http://<server>:8080/` and log in; each gets `users/<name>/challenge` on the server.
- `--progress-repo DIR` commits each learner's progress to a git clone and pushes it; `--data DIR` moves accounts and folders.
- Admin mode never applies to `serve`, so learners cannot unlock it. Use a trusted LAN: it is plain http.

5 Files that matter

`.academy-data/`	progress (`progress.json`), admin hash. Delete `progress.json` to start over.
`curriculum/dojos.yaml`	belts and their modules, in order
`…/challenges/<id>/`	`challenge.yaml`, `files/` (what learners get), `solution/` (instructor only)

6 Ship a copy to learners

```
./scripts/pack_portable.sh        # → dist/*.tar.gz
```

Solutions are left out unless `INCLUDE_SOLUTIONS=1`. The admin hash is copied in, so learner copies stay locked.